SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Design & Analysis of Algorithms: Greedy, Dynamic & NP-Completeness

Department: CSE	**Course / Subject:** Design & Analysis of Algorithms (CS501PC)
Semester: Semester 5	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Asymptotic notations, Divide-and-Conquer, Greedy Strategy, Dynamic Programming, Backtracking, Branch-and-Bound, and NP-Hard/NP-Complete classification.

Unit 1: Foundations & Divide-and-Conquer

Order of growth (O, Omega, Theta), Recurrence relations, Master Theorem, Merge Sort, Quick Sort randomized analysis, Strassen's matrix multiplication.

Unit 2: Greedy Method

Greedy choice property, Fractional Knapsack, Huffman coding data compression, Job sequencing with deadlines, and Minimum Spanning Trees.

Unit 3: Dynamic Programming

Overlapping subproblems, 0/1 Knapsack problem, Longest Common Subsequence (LCS), Matrix Chain Multiplication, Floyd-Warshall all-pairs shortest paths.

Unit 4: Backtracking & Branch-and-Bound

N-Queens problem, Subset sum problem, Graph coloring, Hamiltonian cycles, Traveling Salesperson Problem (TSP) using 15-puzzle branch and bound.

Unit 5: NP-Completeness

Deterministic vs Non-deterministic algorithms, Classes P, NP, NP-Hard, NP-Complete, Cook's Theorem, and polynomial-time reduction proofs.